Going Down - Game Design Document

Contents

Working Title

Going Down

Progress Videos

Prototype - https://www.youtube.com/watch?v=BETC2vcZ-uA

Milestone 1 - https://www.youtube.com/watch?v=OP69_R-tdU8

Current Status

The game is currently in pre-production, with the initial design phase of the game complete. We have a technical prototype of the shadow detection system working for lights. We have also added some interactable lights that would be used to create puzzles like, glow sticks, movable spotlights, flares etc and have a feature showcase demo map.

Concept Statement

A first-person psychological puzzle game where standing in shadows can kill you. Escape through surreal office-world chambers using light manipulation and temporal powers to return to reality.

Genre(s)

The primary genre of this game is a first person puzzle game, while secondary genres can be psychological thriller and escape room. Some sub genres based on the mechanics can be physics puzzle or environmental storytelling. Games that are comparable to this in some way are Portal, for its puzzle structure and Lightmatter for its similar shadow mechanics

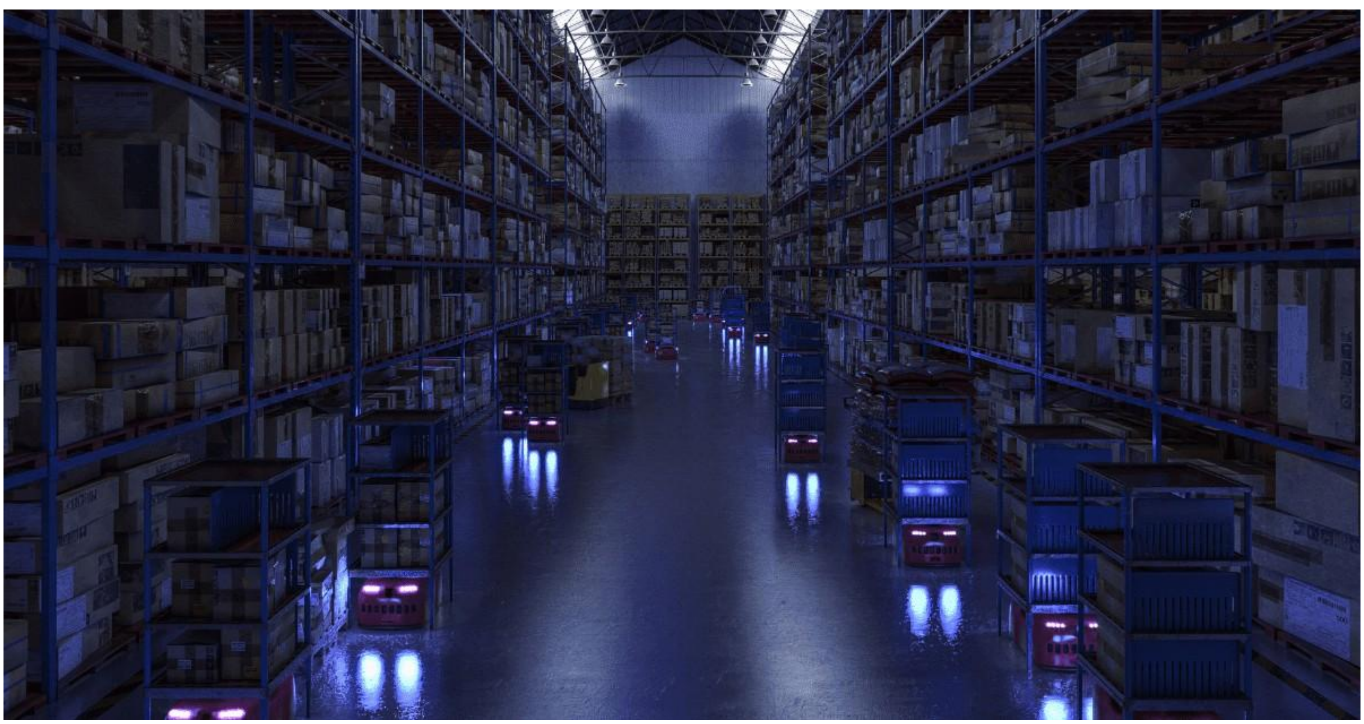
Target Audience

Main Demographic for this game is puzzle game enthusiasts, with age ranges between 18-35. Secondary demographics would be player who enjoy indie games with atmospheric experiences. Its a game people familiar with first-person games, enjoys Portal/Talos Principle. The ESRB rating for this game would be T for Teen for mild supernatural themes, psychological tension.

Gameplay Overview

This places players in an impossible situation: you're an office worker leaving work but you find yourself trapped in surreal puzzle chambers where shadows literally consume organic matter, and your only escape tool is staying in the light. The core gameplay loop centers around these spatial puzzle solving using mechanics involving lights: navigating through shadow zones (danger zones) by staying in light (safe zones).

The player has to progress through five distinct environments—from familiar office spaces to industrial warehouses each level connected through a elevator shaft—each introducing new complexity layers while maintaining the core shadow-avoidance survival mechanic. The experience escalates from methodical puzzle-solving (planning safe routes, interacting with object/moving lights) to panic-inducing action sequences (sprinting through facilities while shadows chase behind).



The game's setting evolves from mundane corporate environments with a physics-defying puzzle chambers, unified together by the 'mysterious elevator' system that is used as both narrative device and level transition (idea from the game 'cocoons' level transition). The visual style would emphasize a dramatic chiaroscuro lighting , which is a stark contrasts between safe the brightly-lit zones and deadly pure-black shadows—creating an atmosphere that's simultaneously otherworldly and make it easy for the player to recognize what's safe and what's dangerous.

Core Loops

Micro Loop (30 seconds):

Observe room layout → identify shadow patterns → plan safe movement path → execute movement + move lights → adapt if plan fails

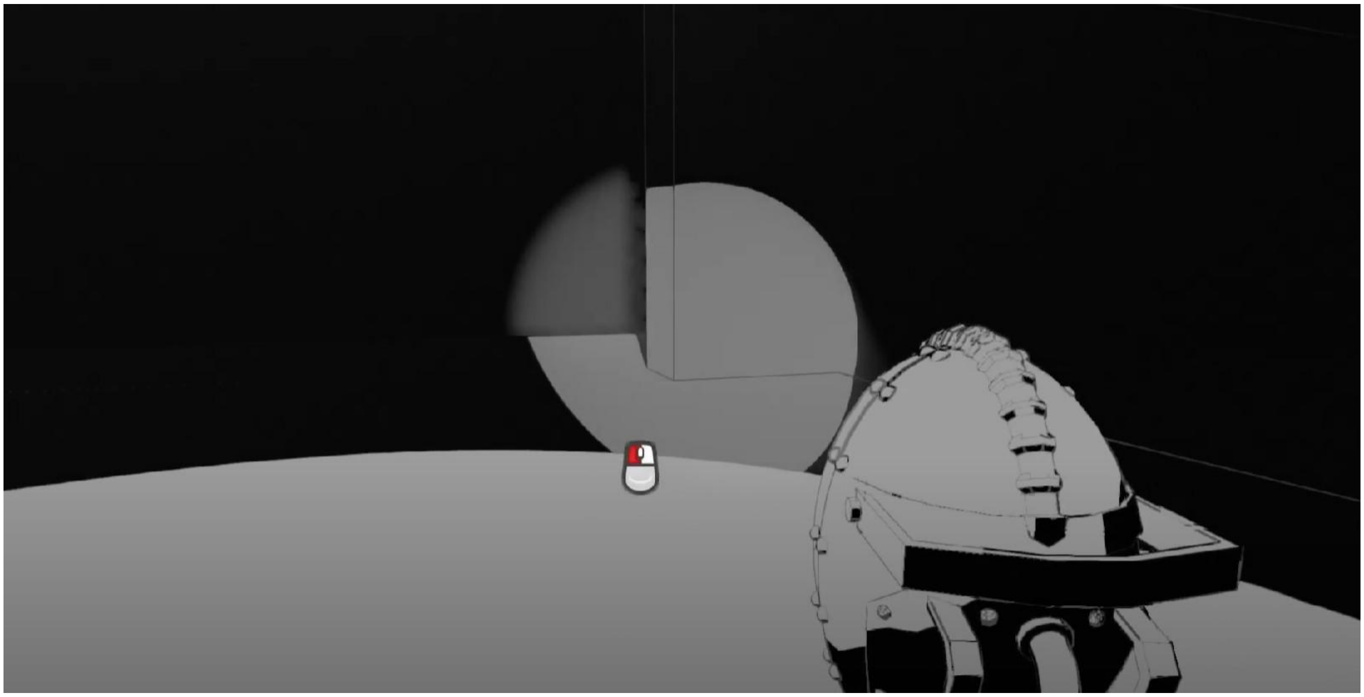
Macro Loop (3 minutes):

Enter new room via elevator → learn new environmental challenges → master increasingly complex mechanic combinations → solve room's central puzzle → progress to next elevator

Unique Selling Points

1. Dynamic Physics-Based Safe Zones

What makes it unique: Unlike traditional collision-based safe zones, my lighting system would real-time raycast to detect for safe areas for the player that precisely match visible lighting (just to make it pixel perfect). When a spotlight rotates/moves, the safe zone dynamically changes in real-time. When objects block light sources, shadows appear exactly where players expect them. This creates unparalleled immersion where the visual lighting and gameplay mechanics are perfectly synchronized.



2. Genre Transformation

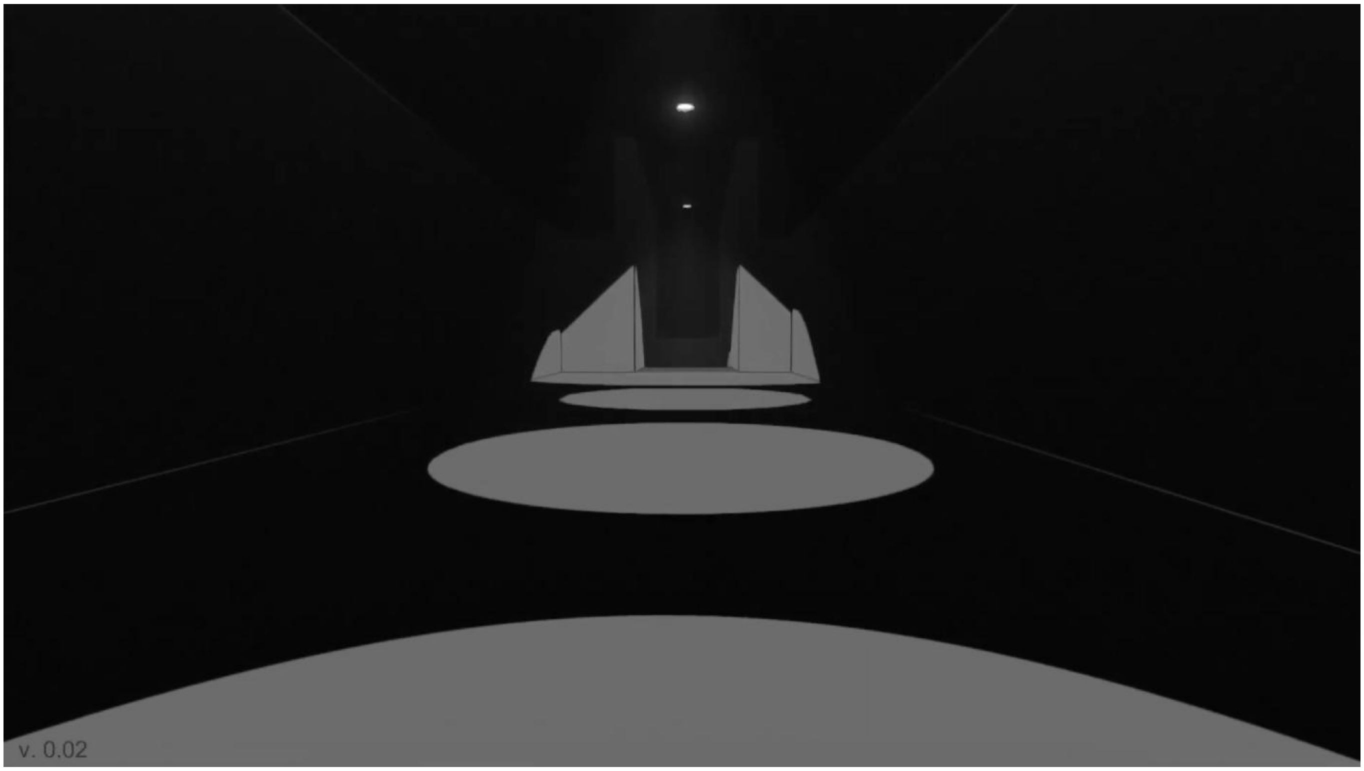
What makes it unique: This game transforms from contemplative puzzle game to panicinducing action thriller within a single 15-minute demo. The same mechanics that enable careful planning become survival tools during the boss chase sequence. This genre evolution—puzzle to action while maintaining identical mechanics—creates a gameplay arc unmatched by staticgenre games.

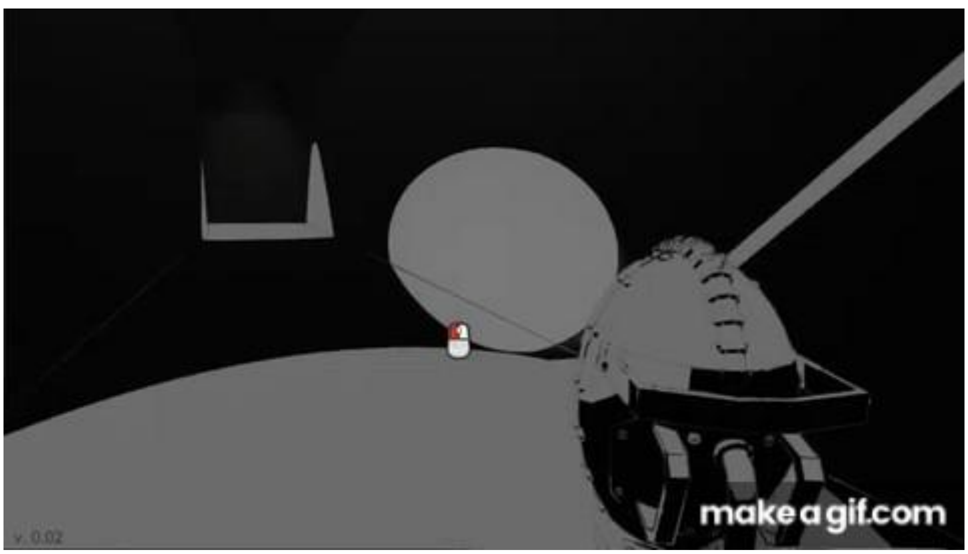
3. Environmental Narrative Integration

What makes it unique: The game tells its story entirely through environmental details without breaking immersion with any cutscenes, dialogue, or UI tutorials. Players learn mechanics by discovering scrawled warnings ("AVOID THE SHADOWS"), abandoned tools, and mysterious notes. The narrative unfolds through spatial exploration and environmental observation, making story discovery feel earned rather than delivered.

4. Psychological Liminal Space Design

What makes it unique: The mundane-to-surreal-to-mundane narrative structure creates a psychological experience that lingers beyond gameplay. Starting in a familiar office, transitioning through impossible puzzle chambers, and returning to normal reality creates an ambiguous ending that sparks discussion and interpretation. The liminal space concept—existing between normal and abnormal—is rarely explored in gaming with this level of intentionality.





Player Experience and Game POV

Player Identity

The player is an unnamed office worker leaving late at night, transported to impossible puzzle chambers through a malfunctioning elevator system.

Core Fantasy

Escape artist with supernatural abilities solving physics-defying puzzles using light and time manipulation.

Emotional Journey

- **Opening (0-2 min)**: Curiosity, mild unease in familiar office setting
- **Early Game (2-8 min)**: Discovery, satisfaction from mastering mechanics
- **Mid Game (8-12 min)**: Focus, flow state, increasing complexity, moving light and lasers introduced
- **Boss Sequence (12-15 min)**: Panic, adrenaline rush, desperation
- **Ending (15 min)**: Relief, contemplation, ambiguous resolution

Player Engagement Factors

- **Mechanical Mastery**: Learning to combine shadow navigation
- **Environmental Mystery**: Discovering story through world details
- **Escalating Challenge**: Each room introduces new complexity layers
- **Moment-to-Moment Discovery**: Finding creative solutions to spatial puzzles

Key Moments

1. Shadow Discovery (Minute 2)

Player steps into shadow, screen darkens, drowning effect begins. Retreat to light, realize shadows are deadly. Sets core survival rule.

2. Mechanic Combination (Minute 6)

Movable spotlights can be used to light up hallways, Player discovers interactable, can use a glow stick to walk in shadows or use a flares to light up areas.

3.Mirrors and Glass (Minute 7)

Player discover a new type of light aka

4. Dynamic Light Challenge (Minute 9)

Rotating spotlight creates shifting safe zones. Player must time movement with light patterns. Introduces urgency. Some lights flicker, or move that the player must navigate through

5. Boss Sprint (Minute 12)

First sprint sequence. Lights failing behind player, must use interactable or push doors while shadows chase. Panic peak.

6. Final Escape (Minute 14)

Elevator doors closing as shadows consume everything. Last-second escape to normal world. Relief and ambiguity.

Art, Sound and Music

Visual Style

- **Art Direction**: Realistic with subtle stylization using Unreal Engine 5's natural rendering
- **Color Palette**:
  - Base: Muted office colors (beiges, grays, corporate blues)
  - Contrast: Dramatic white lighting vs. pure black shadows
  - Accent: Warm yellows/oranges for safe light sources
- **Lighting Design**: High contrast chiaroscuro emphasizing light/shadow boundaries
- **Environmental Design**: Liminal office spaces transitioning to abstract puzzle chambers
- **Dying Effect**: When you start drowning in the shadow the screen has a dying effect and slowly becomes all black.

Audio Design

- **Ambient Audio**:
  - Fluorescent light buzzing (creates unease)
  - Distant mechanical hums
  - Electrical crackling from broken systems
- **Shadow Audio**:
  - Muffled/underwater audio when in shadows
  - Rising drone indicating danger
  - Sharp cutoff when returning to light
- **Music**:
  - Minimal ambient compositions
  - Tension building during complex puzzles
  - Probably some intense electronic music during boss chase
  - Silence in contemplative moments

Reference Materials

- **Visual**: Lightmatter's clean aesthetic, The Stanley Parable's office environments
- **Audio**: Portal's mechanical ambience, Limbo's minimalist sound design
- **Lighting**: Film noir cinematography, architectural photography

Game World Fiction

Setting

A seemingly normal office building that connects to impossible puzzle chambers through its elevator system. The space exists in a liminal state between reality and otherworld.

Narrative Structure

- **Act I**: Leaving work late, elevator malfunction, arrival in puzzle world
- **Act II**: Progressive discovery of shadow mechanics through increasingly complex chambers
- **Act III**: Systemic collapse, escape sequence, return to normal reality

  Resolution: Ambiguous ending - dream, parallel dimension, or psychological episode up to the player to think?

Environmental Storytelling

- **Office Details**: Personal belongings, work schedules, corporate motivational posters
- **Transition Clues**: Elevator buttons that don't correspond to building floors
- **Puzzle World**: Impossible architecture, physics-defying spaces, abandoned maintenance logs
- **Mystery Elements**: Who built these chambers? Where do the shadows come from? Previous visitors? Is this even real? is this a dream?

Lore Elements

- Scrawled warnings: "AVOID THE SHADOWS" in desperate handwriting
- Environmental hints: Certain repeated objects like moving spotlights, mirrors, lasers etc.
- Abandoned tools: Failed repair attempts using normal methods

Current Target Platform

Primary Platform: PC (Windows, Mac, Linux)

Engine: Unreal Engine 5
System Requirements:

- **Minimum**: GTX 1060 / RX 580, 8GB RAM, i5-8400 / Ryzen 5 2600
- **Recommended**: RTX 3060 / RX 6600, 16GB RAM, i7-10700 / Ryzen 7 3700X
- **Storage**: 8GB available space

Notes: Recasting to light might need some optimizations in order to be performant, but rest of the requirements were copied from games like Superluminal and The Stanley Parable which were both made in UE with similar gameplay.

Distribution: Steam (primary), itch.io (secondary)

Future Platforms: Console versions if PC launch successful, but probably not.

Competition Analysis

Direct Competitors

1. **Lightmatter (2020)**
   - **Similarities**: Shadow avoidance mechanic, first-person puzzles
   - **Differences**: We added more light based elements, shorter experience, office setting
   - **Revenue**: ~$100K estimated based on Steam reviews
   - **Our Advantage**: Unique mechanic combination, more focused scope
2. **Portal 2 (2011)**
   - **Similarities**: First-person puzzle structure, physics-based solutions
   - **Differences**: Different core mechanics, shorter experience, indie scope
   - **Revenue**: $200M+ (not directly comparable due to AAA scope can not beat that)
   - **Our Advantage**: Fresh mechanics, modern production values

Market Position

Indie puzzle game filling gap between Lightmatter and Stanley Parable by combining their core mechanics in focused 15-minute experience.

Monetization Strategy

Revenue Model

- **Price Point**: $9.99 USD (comparable to similar indie puzzlers)
- **Justification**: Focused, high-quality experience with unique mechanics

Post-Launch Strategy

- **Content Updates**: Additional rooms, more story and backstory about the main character if successful
- **Community**: Steam Workshop support for user-generated puzzles, since only one main mechanics is needed to make a puzzle and then just a start and goal.

Player Objectives and Progression

Primary Objective:

Escape the puzzle dimension and return to normal reality

Room-by-Room Goals:

Room 0: Office (Tutorial) - 2 minutes

- **Learn**: Shadow avoidance basics
  Discover: Shadow avoidance
  Master: Basic [E] and [Q] interactions
  Objective: Fix broken lamp, reach elevator

Room 1: Warehouse - 3 minutes

- **Learn**: Moving light sources, timing-based movement
- **Challenge**: Navigate between storage shelves with overhead crane spotlight
- **Mechanics**: Moving lights to get through, activate switches, use flares and glowsticks
- **Objective**: Reach loading dock elevator

Room 2: Server Room - 3 minutes

- **Learn**: Multiple simultaneous objects, complex timing
- **Challenge**: Maze of server racks with flickering lights
- **Mechanics**: Repair server cooling systems, reroute power through laser using mirrors or glass
- **Objective**: Access core server elevator

Room 3: Parking Garage - 3 minutes

- **Learn**: Vertical navigation, car headlight manipulation
- **Challenge**: Multi-level space with ramps and platforms
- **Mechanics**: Repair generator to turn on lights, turn car headlights on vehicles
- **Objective**: Reach maintenance elevator to building systems

Boss: Elevator Shaft - 4 minutes

- **Challenge**: Sprint sequence with systems failing
- **Mechanics**: All previous mechanics under time pressure
- **Phases**:
  - Sprint with lights failing behind (1 min)
  - Broken door repair while being chased (1 min)
  - Obstacle gauntlet with switches (2 min)
- **Objective**: Reach final elevator before shadows consume everything

Skill Progression

- **Rooms 0-1**: Learn individual mechanics
- **Rooms 2-3**: Combine mechanics creatively
- **Boss**: Master mechanics under extreme pressure

Pacing Design

- **3-4 minute room cadence** maintains engagement
- **Elevator transitions** provide narrative beats and anticipation
- **Complexity curve** ensures players never feel overwhelmed
- **Boss escalation** transforms contemplative puzzles into action sequence

Game World Layout

Note: Puzzles here might change based on how the mechanics are actually implemented and nothing about the actual puzzle in each room is set in stone.

Room 0: Corporate Office

- **Size**: small open office space
- **Lighting**: Standard fluorescent overheads (2 broken, 1 working)
- **Shadows**: Desk areas, under broken fixtures
- **Puzzles**: Single broken lamp blocks path to elevator
- **Props**: Desks, computers, motivational posters, coffee cups
- **Atmosphere**: Familiar, slightly unsettling late-night office vibe

Room 1: Industrial Warehouse

- **Size**: big room with a high ceiling
- **Lighting**: Overhead industrial fixtures on moving crane
- **Shadows**: Between 4m tall storage shelving units
- **Puzzles**: Navigate rotating spotlight, repair conveyor belt
- **Props**: Pallets, forklifts, shipping containers, control panels
- **Atmosphere**: Industrial, mechanical sounds, larger scale

Room 2: Server/Data Center

- **Size**: maze-like layout with server racks
- **Lighting**: Individual server rack lights, emergency lighting
- **Shadows**: Corridors between 2m tall server banks
- **Puzzles**: Repair cooling systems, reroute power through servers
- **Props**: Server racks, cables, monitoring displays, cooling units
- **Atmosphere**: Electronic humming, blinking LEDs, digital aesthetic

Room 3: Underground Parking

Size: multi-level (2 floors connected by ramps), biggest room

Lighting: Car headlights, parking structure fixtures

Shadows: Behind concrete pillars, vehicle shadows

- **Puzzles**: Use car headlights as light sources, repair barriers
- **Props**: Cars, concrete barriers, traffic cones, exit signs
- **Atmosphere**: Echoing space, car alarms, urban underground feel

Boss: Vertical Elevator Shaft

- **Size**: Long hallways through the a office corridor
- **Lighting**: Emergency lights failing sequentially from bottom up
- **Shadows**: Rising darkness chasing player upward
- **Puzzles**: platforms, doors, switches while climbing
- **Props**: Industrial piping, maintenance platforms, electrical panels
- **Atmosphere**: Mechanical failure, alarms, rising tension

User Interface Design

Controls:

- Movement: WASD and mouse movement + space to jump
- Interact: E pick/interact or Q to use

Core Philosophy:

Minimal UI, Maximum Immersion

- **No HUD elements** during gameplay
- **No tutorials** - learning through environmental cues
- **No health bars** - audio/visual feedback indicates shadow danger
- **No menus** during play - pure first-person experience

Visual Feedback Systems

- **Object Highlighting**: Subtle glow when targeted
  - Interactive objects: White edge glow
- **Reusable Assets**: Player will learn to recognize object like switches, elevators by constantly using them in different level. Think of like how the yellow ladder in resident evil works
- **Shadow Warning**: Screen edges darken when near deadly shadows and some sound ques too.

Audio Feedback

- **Interaction Success**: Satisfying click/activation sound, (like elevator ding when level complete)
- **Shadow Danger**: Underwater/muffled audio effect
- **Environmental**: 3D positional audio for all game elements

Screen Flow

1. **Main Menu**: Start Game, Settings, Quit
2. **In-Game**: Pure first-person view, no overlays
3. **Pause Menu**: Resume, Settings, Quit to Menu
4. **End Screen**: Completion time, replay option

MVP Systems and Features

1. Shadow Detection System

Intent: Create precise, performance-friendly detection of player safety based on realistic lighting

Implementation:

- **Raycast To Player Feet**: Cast a ray from the player feet to the
- **Dynamic Mesh Generation**: Connect raycasts hit points to form collision boundaries
- **Collision Detection**: Use generated meshes for overlap events
- **Performance**: Cache static light meshes, only recalculate for moving lights

Edge Cases:

- **Multiple overlapping lights**: Union of safe zones
- **Moving lights**: Recalculate mesh every 0.1 seconds
- **Blocked lights**: Raycast determines actual coverage area
- **Performance scaling**: Reduce ray count for distant lights

2. First-Person Controller

Intent: Responsive movement system optimized for precise puzzle navigation

Features:

- **Standard FPS movement**: WASD + mouse look, plus sprint (unreal has a player movement system)

  Jump mechanics: Standard gravity with coyote time

  Interaction range: 4 meter reach for object targeting

  Smooth camera: No motion sickness, stable framing

3. Lighting System

Intent: Support both atmospheric lighting and gameplay-critical light sources

Light Types:

- **SafeLights**: Generate collision meshes, protect from shadows
- **Ambient Lights**: Visual only, no gameplay impact
- **Moving Lights**: Update collision meshes dynamically

Performance Targets:

- **Light Count**: Up to 10 SafeLights per room, will depend on performance
- **Shadow Quality**: Medium shadows for performance
- **Update Frequency**: Moving lights recalculated 10fps

4. Audio System

Intent: Provide clear gameplay feedback and atmospheric immersion

Audio Categories:

- **Gameplay Audio**: Interaction feedback, danger warnings
- **Atmospheric Audio**: Environmental ambience, spatial audio
- **Music**: Dynamic scoring responding to tension level
- **Voice**: Minimal, only environmental story elements

Technical Implementation:

- **3D Audio**: Full spatial positioning for all sources (unreal does this already)
- **Dynamic Mixing**: Muffle audio when in shadows
- **Performance**: Audio LOD system for distant sources

Game Objects

Environmental Objects

Lights (SafeLight)

- **Function**: Generate safe zones, can be broken/repaired
- **States**: Working (safe zone active), Broken (no protection), Repairing (transition)
- **Types**: Desk lamps, ceiling fixtures, car headlights, emergency lighting
- **Audio**: Electrical hum when working, crackling when broken

Doors

- **Function**: Block progress until opened/repaired
- **States**: Open, Closed, Broken, Repairing
- **Interactions**: [E] to open, [I] to repair if broken
- **Audio**: Mechanical sounds, door closing/opening

Switches

- **Function**: Control lights, platforms, other machinery
- **States**: On, Off
- **Interactions**: [E] to toggle
- **Visual**: Clear on/off indication through LED or position

Placeable / Movable lights

- **Function**: Player controlled light - lasers, spotlight, flares and glowsticks
- **States**: On, Off
- **Interactions**: [E] to toggle [Q] to use or place
- **Visual**: Clear animation or noise

Interactive Systems

Elevators

- **Function**: Transition between rooms, narrative pacing
- **Interactions**: [E] to call/enter elevator
- **Audio**: Mechanical elevator sounds, music stingers
- **Visual**: Floor indicators, opening/closing doors

Moving Platforms

- **Function**: Create timing-based challenges with light/shadow
- **States**: Active (moving), Off (stationary)
- **Interactions**: Switch-controlled or automatic
- **Timing**: Predictable patterns for player planning

Hazard Objects

Shadow Zones

- **Function**: Kill player on contact, create navigation challenges
- **Implementation**: Absence of SafeLight collision volumes
- **Effects**: Screen darkening, audio muffling, health drain
- **Visual**: Pure black areas with particle effects at edges

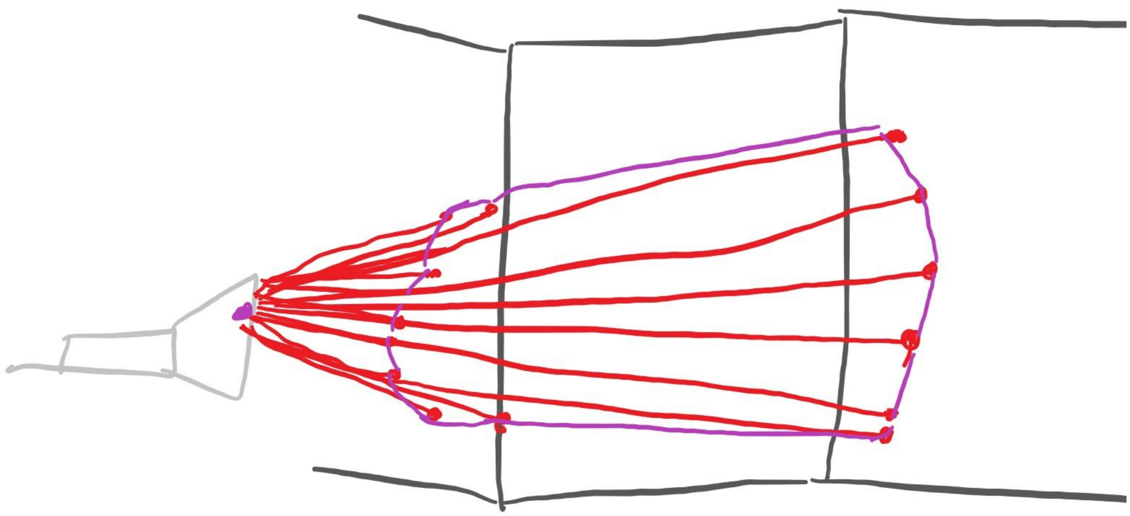
Technical Documentation

Major Technical Challenges

1. Dynamic Collision Mesh Generation

- **Challenge**: Real-time creation of collision volumes based on lighting
- **Solution**: Raycast-based mesh generation with performance caching
- **Fallback**: Pre-placed collision volumes if performance insufficient
- **Testing**: Profile with 10+ lights, maintain 60fps minimum

Simple drawing how the mesh will look like when raycast point are joined back, then the area of intersection is used as a safe area.



2. Mirror reflections and Glass refractions

- **Challenge**: Real-time reflection and refractions light rays based on material and hit
- **Solution**: Raycast-based path making Realtime
- **Fallback**: Pre-determined paths for light
- **Testing**: Profile with 10+ reflections and refractions, maintain 60fps minimum

Development Schedule and Milestones

Prototype (Completed - Submitted)

Core Mechanics

- First-person movement (WASD + jump + sprint)
- Shadow detection system using raycast to lights
- Cone angle detection for spotlights using dot product math
- Point light radius detection
- Speed penalty when in shadows (half speed)
- 1-second death timer in shadows
- Death and respawn system
- Jump makes player temporarily safe from shadows

Light Systems

- BP_SafeLight_Spot and BP_SafeLight_Point with BPI_SafeLight interface
- Sphere collision detection for prototype
- Toggle lights on/off via interaction
- Lights register with BP_LightManager
- ActiveLights array tracking

Game Loop

- Interaction system (E key)
- Win condition (door interaction ends game)
- Basic vignette effect (Material Parameter Collection) - Incomplete

Level Design

- Dev room with dramatic chiaroscuro lighting
- Multiple spotlights creating shadow gaps

Milestone 1

Visual Polish

- Basic vignette effect (Material Parameter Collection) - completed
- Death sequence with drowning effect
- Perlin noise shader for screen distortion
- Improved vignette with noise texture
- Shadow catcher shader (wavy/glossy effect in dark areas)
- Cel-shading for environment
- Simple map with all mechanics showed

Light Mechanics

- **Carriable spotlights** - pick up, place in look direction
- **Glow sticks** - pick up, crack (Q), temporary light with duration timer
- **Flare/Emergency light** - throwable, lights area temporarily
- **Flickering lights** - on/off cycle pattern
- **Proximity lights** - turn off/on when player near trigger volume
- Edge case testing for all light tools
- Dynamic light testing with current detection system
- Performance check

Core Systems

- **Checkpoint system** - track last checkpoint per map, respawn at checkpoint
- **Elevator spawn/exit** - spawn in elevator (doors open), exit elevator (interact button, doors close)
- **Inventory system** - scroll wheel to cycle items, no on-screen UI
  - Hold 1 glow stick max
  - Hold 1 flare max
  - Q to use, E to drop
  - Movement restrictions when holding items (no sprint/jump)

Item Rules

- Items drop on death (stay at death location)
- Checkpoints restock consumables
- Can't pick up duplicates if slot full

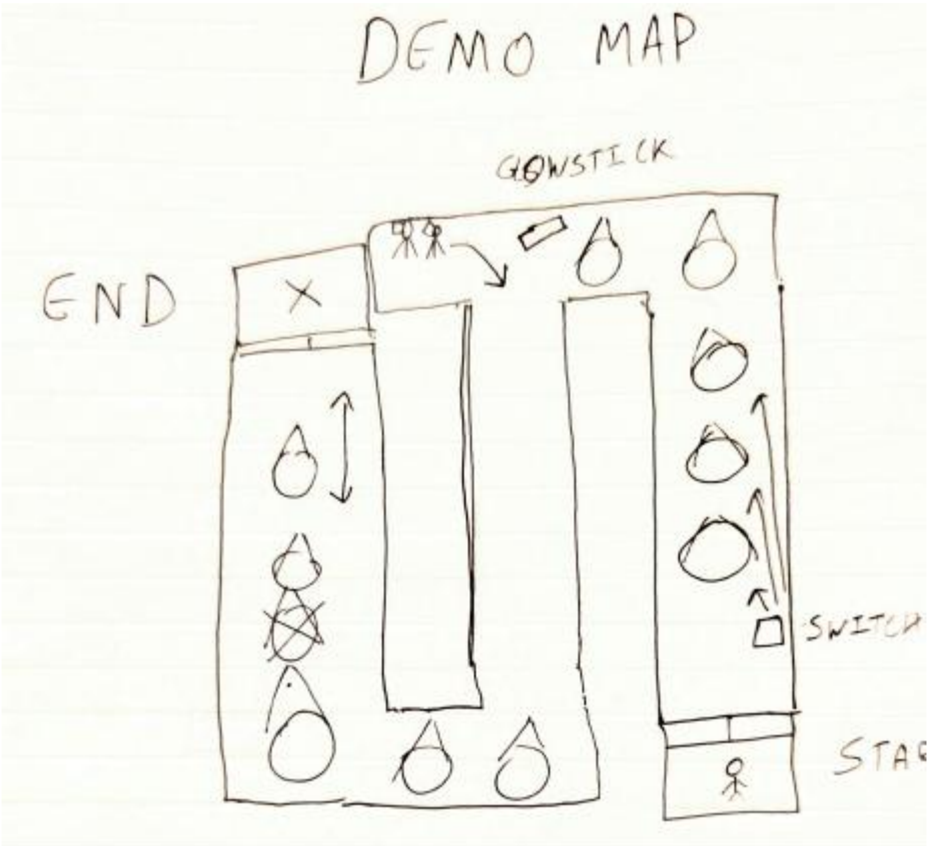
Uncomplete Tasks

- **Inventory system** - scroll wheel to cycle items, no on-screen UI
    - Hold 1 glow stick max
    - Hold 1 flare max
    - Q to use, E to drop
    - Movement restrictions when holding items (no sprint/jump)
- Some polish for pickup and temporary lights
- Death sequence with drowning effect
- Perlin noise shader for screen distortion
- Improved vignette with noise texture
- Shadow catcher shader (wavy/glossy effect in dark areas)

Bugs and Issues

- Light placement might be off from the floor cause of the line trace not hitting the floor
- Light can be placed inside objects right now
- Flare has some issue with physics and not being able to hit the light collider
- Flare throw might need to know where the flare lands for puzzles
- Flare Light might be inside the floor sometimes when thrown

Demo Map



Milestone 2

Inventory + Item Handling

- Scroll wheel to cycle items (no UI)
- Hold limits:
  - 1 glow stick max
  - 1 flare max
- Actions:
  - Q = use
  - E = drop
- Items drop on death
- Checkpoints restock
- Display small text prompt: “Q to Use”
- Movement restrictions when holding items (no sprint/jump)
- Full edge-case coverage:
- Death holding item
- Picking up item with slot full
- Dropping during animations

- Pickup of glowstick already cracked
  Item Fixes
- Flare physics issue (not colliding with light collider)
- Flare detection properly counts as light
- Moveable spotlight placement:
- Fix line trace floor hit
- Prevent placement inside objects
- Only place when valid

Light System Optimization

- Reorder active lights array by distance
- Limit raycasts per frame
- Fix flare light position not sinking into floor

Level/World Systems

- Level transitions using elevators
- Enter elevator → doors close → load next room
- Spawn in elevator → doors open
- Checkpoints fully working across multiple rooms
- Basic level flow foundation

Quality & Stability

- Basic polish on glow stick, flare visuals
- Temp items look good when dropped
- SFX placeholders (optional)

- **Light ray/laser beam** - thin cylinder with emissive material
  - Generate on-the-fly by scaling mesh and collider to line trace hit distance
  - Box collider along path = safe zone
  - Not real light, just visual + collision detection
- **Mirrors** - reflect light rays following law of reflection
  - Spawn reflected ray at bounce angle
  - Moveable/rotatable mirrors
- **Refraction** - light rays bend through glass using Snell's Law
  - Glass windows, fish tanks, water dispensers
  - Double refraction (entering and exiting surface)
  - Total internal reflection at steep angles
- **Prism to point light** - laser hits prism, converts to omni-directional point light

Optional: Visual Improvement

- Some polish for pickup and temporary lights
- Death sequence with drowning effect
- Perlin noise shader for screen distortion
- Improved vignette with noise texture
- Shadow catcher shader (wavy/glossy effect in dark areas)

Milestone 3

New Light Tools

- **Light ray/laser beam** system
  - Dynamic cylinder mesh
  - Collision box safe zone
  - Auto-scaling using line trace
- **Mirrors**
  - Reflect laser with law of reflection
  - Moveable / rotatable
- **Refraction through glass**
  - Snell's Law
  - Double refraction
  - Total internal reflection
- **Prism system**
  - Convert laser → point light

Puzzle Mechanics

- Pressure plate → activates light/door
- Timed doors
- Battery swapping
- Keypad codes
- Simple lock + key system

Milestone 4 and 5

Level Building

- Build all puzzle rooms (approx 10–15 min playtime)
- Add transitions between all rooms
- Environmental storytelling
- Visual polish passes

Gameplay Polish

- Polish all existing mechanics
- Performance optimization
- Lighting balance
- QA & playtesting

Audio

- Footsteps
- Light hum
- Glow stick crack
- Flare fizz
- Elevator sounds
- Death audio

Final Testing & Balancing

- Difficulty curve
- Consistency of shadow zones
- Ensure performance stable with multiple lasers

Unresolved Questions

Current Design Questions

1. **Room Count**: Keep 4 rooms + boss, or expand to 6 rooms for longer experience?
   - **Decision Date**: End of Month 4 after playtesting
   - **Criteria**: Pacing feedback, development timeline
2. **Difficulty Scaling**: How steep should the learning curve be?
   - **Decision Date**: Month 6 during playtesting phase
   - **Criteria**: Player success rates, frustration points

Resolved Questions

- **UI Approach**: Minimal UI confirmed, no HUD elements
- **Control Scheme**: [E] interact, [Q] to use the interactab
- **Art Style**: Realistic Unreal Engine look confirmed
-

Platform Priority: PC first, console later if successful

Ideas and Expansions

Post-MVP Features

- **Additional Rooms**: Cafeteria, maintenance tunnels, rooftop
- **New Mechanics**: Add "things" crawl out of shadows and player has to use a flashlight to kill them, flashlight takes in a battery that drains so use it accordingly
- **Community Features**: Steam release, Level editor, Steam Workshop integration

Sequel/Expansion Concepts

- **Different Setting**: Hospital, school, shopping mall with same mechanics
- **Co-op Mode**: Two players coordinating light and time powers
- **Longer Experience**: 2-3 hour game with deeper puzzle complexity

Changelog

10/19/2025

- Removed the time-reversal mechanic and instead focused on perfecting light mechanic and adding on top of it
- Add more mechanics involving lights, like - mirror reflection, water/glass refraction, light making interactable like flares and glow sticks that create temporary light and are a limited resource.
- Add the goals and features that have been implemented in the prototype and the future goals for milestone 1 and milestone 2

11/14/2025

- Updated the new scope for Milestone 2 to include some backlog task and certain things missed in milestone 1 for feature polish
- Moved certain non functional tasks out of milestone 2 and save them for future milestones
- Added Links for video demos of Prototype and Milestone 1
- Added the current demo map

11/30/2025

- Rescoped Milestone 2 to focus exclusively on completing core systems and stabilizing item/light mechanics (since I was having a lot of issues with physics and placements with current structure of the code).

- Moved all **advanced light systems** (laser beams, mirrors, refraction, prisms) to **Milestone 3** to reduce risk and avoid feature overload.
- Added **Level Transition System** (elevator entry/exit) to Milestone 2 to support multi-room progression.
- Added all missing polish from Milestone 1 (flare fixes, spotlight placement validation, item handling edge cases) into Milestone 2 backlog.
- Added the task distribution for milestone 3 and 4.